

Color Seeker

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1. Introduction

Autonomous robots are becoming increasingly important in environments where manual navigation and searching are inefficient, dangerous, or impossible for humans. Applications such as search and rescue, warehouse automation and inspection systems all rely on robots being able to independently explore unfamiliar environments while identifying important targets. The motivation behind this project was to create a robotic system capable of actively searching an unknown environment while adapting to changing goals in real time. Rather than following a fixed route, the robot needs to explore autonomously, building an understanding of its surroundings, and remembering the locations of previously discovered targets for future use.

To accomplish this, we developed a ROS2 based robotic simulation using a TurtleBot3 Burger robot inside a Gazebo house environment. The robot was designed to navigate through the environment, detect colored cubes using a camera system, and store their locations once discovered. The system can then switch between different target colors and either continue exploring to locate new targets or revisit previous stored locations. By combining autonomous exploration with object detection, the project demonstrates how robotic systems can intelligently sense, interpret, and interact with dynamic environments without requiring human control.

2. Relevance to Course

This project is highly relevant to CSCI 4552 because it integrates many of the major robotics concepts covered throughout the course into a unified autonomous system built on ROS2. By using the SLAM toolbox for simultaneous localization and mapping, the robot is able to construct and interpret a map of its environment while estimating its own pose in real time. This is combined with the Nav2 navigation stack and explore_lite package, allowing the robot to

autonomously explore unknown environments through frontier exploration, perform path planning, and avoid obstacles while navigating toward goals. The project also incorporates computer vision through OpenCV and Cv_Bridge for real time color detection using camera sensor data from the robot. In addition, the project demonstrates the modular design of ROS2 through the use of publish/subscriber nodes, sensor integration, and communication between multiple subsystems responsible for perception, navigation, mapping, and memory storage. The robot processes information from lidar and camera sensors, uses that information for decision making, and stores previously discovered target locations for future navigation tasks. These components reflect real world robotics architecture where multiple independent systems must work together seamlessly. Overall, the project provided hands-on experience with autonomous exploration, SLAM, robot perception, sensor processing, ROS2 system integration, and intelligent navigation, making it directly applicable to the core learning objectives of the course.

3. Methodology

3.1 Core Details

The project was developed using a simulated robotics environment built with Gazebo and ROS2. The primary robot platform used was the TurtleBot3 Burger, which was placed inside the *turtlebot3_house.world* environment. Colored cubes were manually spawned throughout the environment at set locations to serve as search targets for the robot. A camera sensor was added to the robot model to support real time color detection, while the built in lidar sensor was used for mapping, localization, and obstacle avoidance.

The first stage of the project focused on establishing autonomous navigation and mapping capabilities. The SLAM toolbox was used in the system so the robot could perform simultaneous localization and mapping. As the robot moved through its environment, lidar data was continuously processed to generate a map while estimating the robot's pose within the map. This allowed the robot to operate in an initially unknown environment without relying on any prebuilt maps.

To allow the robot to autonomously explore the environment, the Nav2 navigation stack and `explore_lite` package were implemented. The `explore_lite` node identifies frontiers, which are boundaries between unexplored and explored areas. Once a frontier was selected, Nav2 generated a valid path toward the target while handling path planning and obstacle avoidance. As the robot traveled, new lidar data expanded the map, creating additional frontiers and enabling continuous exploration of the environment. This process repeated until the environment had been sufficiently explored, and the colored cubes were found. There are three config files in the config folder for tweaking the navigation, exploration, and SLAM settings: `explore_params.yaml`, `nav2_params.yaml`, and `slam_params.yaml`. These files carry a lot of the work of adjusting the behavior of the robot to make it as effective as reasonably possible in a given environment.

Computer vision was implemented using OpenCV and `CV_Bridge`. Camera frames published through ROS topics were converted into OpenCV image formats for processing. The system analyzed image data to detect specific target colors corresponding to the colored cubes placed throughout the environment. When a target object was identified, the robot associated the detected object with its current pose. These positions were then stored in memory so the robot could later revisit previously discovered cubes without needing to search through the entire environment again.

An optical character recognition (OCR) system was added to allow the robot to read text from flashcards using its onboard camera feed. This feature was implemented using OpenCV, `CV_Bridge`, and the Tesseract OCR Engine within a dedicated ROS2 node. The node subscribes to live camera images published through a ROS topic, converts the images into an OpenCV format, and processes them at a fixed interval to reduce unnecessary computation. The captured frames are converted into RGB format and passed through the OCR engine to extract any visible text. Additional filtering and regular expression processing were used to remove whitespace, symbols, and nonalphabetical characters before publishing the cleaned text to a separate ROS topic. This system allows the robot to recognize text-based commands from flashcards, creating a more interactive method for selecting a target color for the robot to find.

The final system integrated autonomous exploration, mapping, navigation, computer vision, memory based target tracking and optical character recognition into a single ROS2 architecture using modular nodes and topic based communication. This allowed the robot to seamlessly switch between searching for undiscovered targets and navigating back to already known targets when a different color is selected.

3.2 Work Distribution

Nick Empson was responsible for the initial setup and integration phases of the project. This included configuring the Gazebo environment, spawning the colored cubes throughout the house, and integrating the TurtleBot3 Burger robot into the ROS2 system. Nick also added and configured the onboard camera used for computer vision tasks and implemented the initial waypoint based navigation system used during early testing and development.

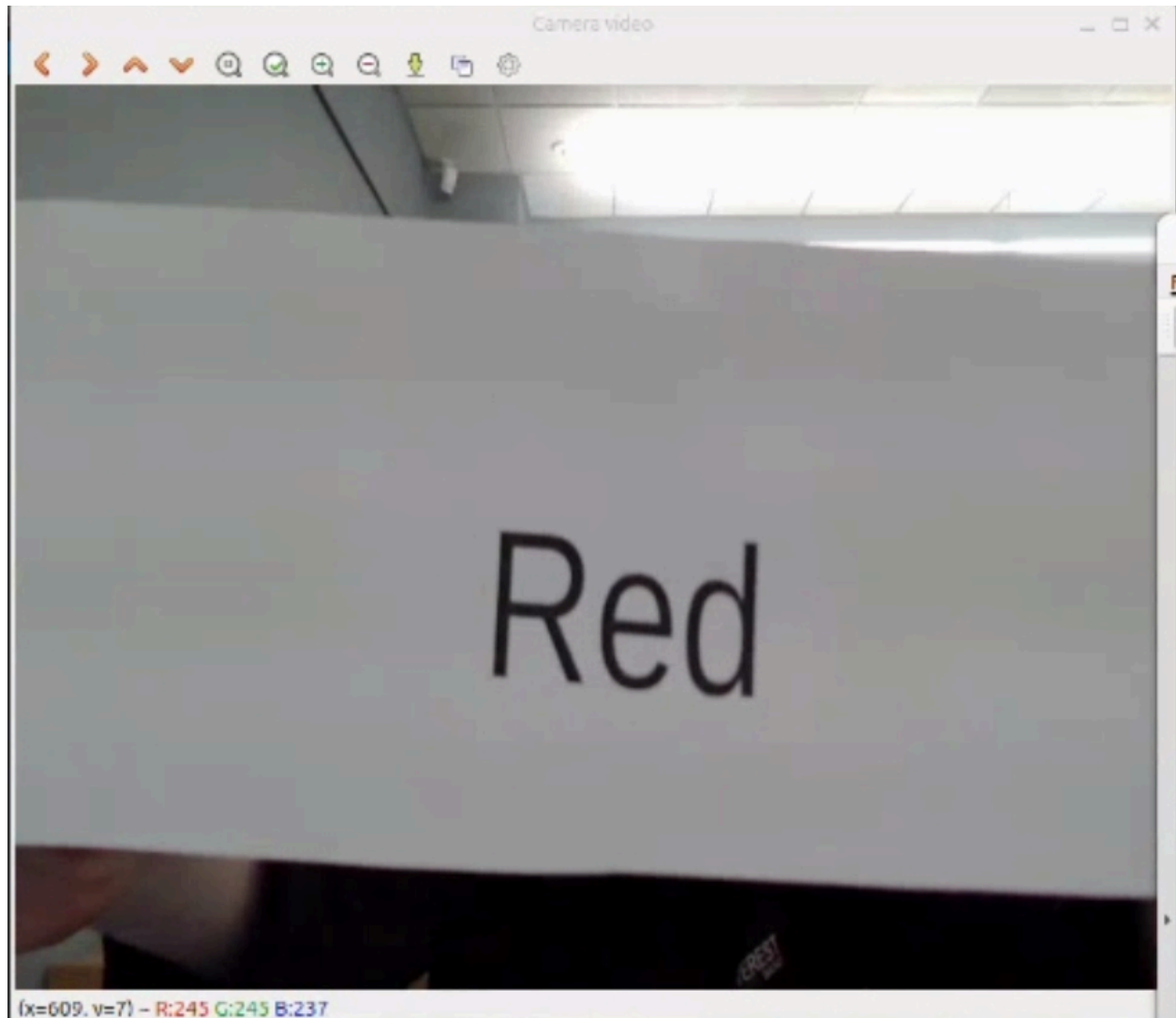
Tyler Hamilton focused on the autonomous navigation and memory-based navigation components of the project. This included integrating the Nav2 navigation stack and the `explore_lite` package to enable frontier exploration of unknown environments. Tyler also implemented the logic for storing previously detected cube locations and allowing the robot to alternate between active exploration and revisiting known target locations depending on the currently selected target goal.

Jason Guoin was primarily responsible for the computer vision and object detection systems. This included processing camera images using OpenCV and implementing the color detection algorithm used to identify the colored cubes throughout the environment. Jason also worked on the live video feed processing system used for detecting flashcards and integrating the OCR text recognition system for a user to choose what target color the robot was supposed to locate.

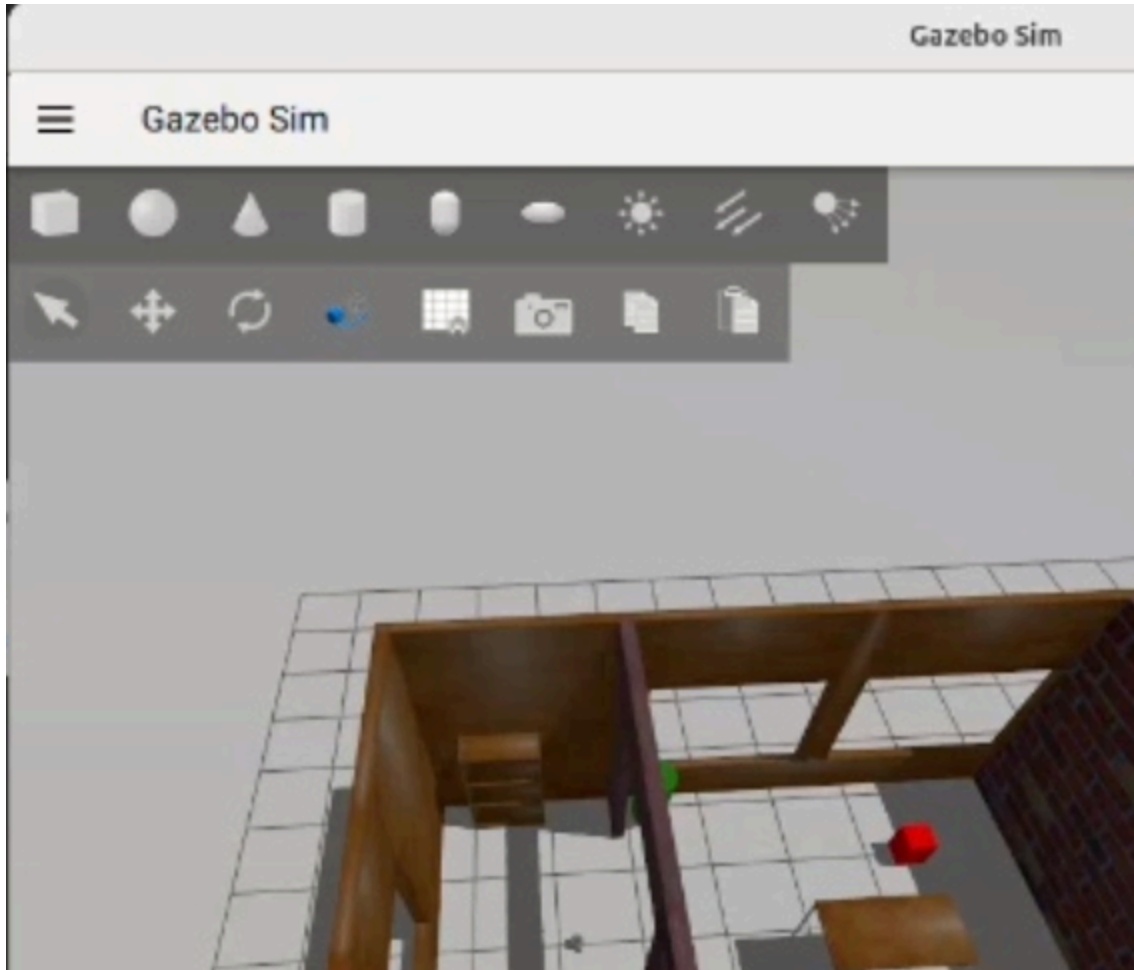
Additional work on project report, project presentation, and video demonstration were evenly divided amongst the team members. Everyone contributed to creating and revising the project requirements. This also included testing and debugging the code.

4. Results

The OCR was functional, but there is likely room for improvement. There were cases such as the one pictured below, where even though the flashcard was quite close to the camera, the text still wasn't being detected. Ideally, the OCR would be better at picking up the text, so that it's easier to use the flashcards.



The color detection worked quite well. Although it was inconsistent whether or not the robot would see the trashcans as green. The trashcans are a different green than the green cube, so it would be reasonable for it to not see it as green, but it's strange that it only sees it sometimes.



Finally, the largest section, the movement. The robot movement works well most of the time. We had to reduce the turning speed to avoid issues where pose drift between the bot's physical and virtual orientations causes ghost walls/obstacles. This slower turning can be seen at around 0:45 in the provided video demonstration. If the bot does bump or crash into anything, the tilts this causes will orient the lidar scanner towards the floor, which will then be detected as blocking walls in front of the robot. To remove some of the ghost obstacles, `observation_persistence` is set to 1.0, which means that if not re-observed in 1 second, the added cost of an obstacle will be cleared. This setting made a huge difference in preventing the accumulation of ghost obstacles over time.

If there is another object along the line of sight between the robot and a colored cube, the robot will sometimes erroneously remember the colored cube's location to be the location of that

object instead. If the location is close enough to the actual location of the cube, then the robot can generally come back to the cube to find it successfully. If it is pretty far away, then the robot will return to the erroneous location, and sometimes can recover and move on after, but sometimes can get stuck. Resolving this would require an additional layer of searching at the destination if the robot returns to the remembered cube location, but cannot visually confirm the presence of that cube with its camera.

The robot does fairly well at exploring the map overall. It achieves its goal of changing the targeted colors through flashcards, searching for colored cubes in the environment, and moving towards them.

5. Conclusion

In conclusion, this project successfully demonstrated the development of an autonomous robotic system capable of exploring an unknown environment, detecting target objects, and navigating using ROS2. By integrating SLAM, Nav2, frontier exploration, computer vision, and OCR based detection into a single system, the robot was able to sense, think, and act with its environment without direct human control. The project provided valuable hands-on experience with real world robotics concepts such as localization, mapping, perception, sensor integration, and autonomous decision making. Additionally, the modular ROS2 architecture allowed multiple subsystems to communicate through topics, services, and nodes, reflecting how modern robotic systems are designed. Overall, the project achieved its primary goals while also creating a strong foundation for future improvements such as more advanced object recognition, expanded environments, and enhanced human-robot interaction capabilities.